

LifeDrop — Blood Donation Network

Complete Technical Project Guide, Newly Added Features & System Architecture

1. Executive Summary & Problem Solved

LifeDrop is a Pakistan-focused voluntary blood donation and hospital emergency coordination network. In Pakistan, acute blood shortages affect over 250,000 thalassemia children, emergency trauma victims, and post-partum mothers annually. The project eliminates middlemen and black marketing by directly matching voluntary donors with patients and hospitals in real time.

Key Technology Stack: Frontend: React 19, TypeScript, Vite, SCSS, React-Router 6, Leaflet Maps, PWA. Backend: Django 6, Django REST Framework (DRF), Daphne ASGI Server, Django Channels (WebSockets), SQLite/PostgreSQL, SimpleJWT Authentication.

2. Newly Added & Upgraded Features (Recent Enhancements)

The following 6 major features were newly designed, developed, and connected to make this project production-grade:

Feature	Module / File	What Was Added & How It Works
1. Hospital Self-Registration & Onboarding	AuthPage.tsx AuthContext.tsx models.py	Role-based signup toggle: Individual Donor vs Hospital/Blood Bank. Collects hospital name, address, 24/7 helpline, and license number. Auto-initializes 8 blood group inventories.
2. Public Hospitals Directory (/hospitals)	HospitalsPage.tsx views.py (HospitalViewSet)	Patients can search verified hospital blood banks across Pakistan. Displays live 8-group stock counts, emergency helplines, 1-click WhatsApp inquiry, and 'Request Blood Here' action.
3. Hospital Blood Bank Command Center	HospitalPanel.tsx views.py (BloodInventoryViewSet)	Interactive reserve management for all 8 blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-). Live +/- unit adjustment buttons, critical low stock alerts (<3 units), and visual progress bars.
4. Emergency External Donor Broadcast	HospitalPanel.tsx request_external_donors	When hospital blood bank is out of stock for an admitted patient, hospital staff broadcasts an emergency request specifying patient MR, blood group, units, and urgency (<2 hours).
5. Community Donors Summon Directory	HospitalPanel.tsx community_donors	Hospital staff can search external verified donors by blood group and city, and directly summon them with 1-click Phone Call or WhatsApp emergency message.
6. WhatsApp 1-Click Normalization Engine	whatsapp.ts SearchBloodPage.tsx	Sanitizes Pakistani local phone numbers (03xx -> 923xx) and formats pre-filled urgent blood request messages that open directly in WhatsApp App and WhatsApp Web.
7. NGO Blood Camps & Drives Page	CampaignsPage.tsx /campaigns route	Directory of upcoming blood donation camps. Multi-city filters (Lahore, Karachi, Islamabad), timings, volunteer registration form, and 'Host a Drive' modal.

8. Digital Certificate of Appreciation	DonationHistory.tsx	Automated LifeSaver award certificate generation for completed donations. Renders donor name, donated volume, date, hospital, gold seal, and Print/Save PDF support.
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3. Total Complete Feature Matrix

A. Donor Discovery & Matching:

- **Smart Compatibility Matching:** Matches both exact blood groups and medically compatible donor groups (e.g. O-universal donor, AB+ universal recipient).
- **Interactive Leaflet Map:** Visualizes nearby donors across Pakistani cities with distance calculation (km).
- **90-Day Medical Cooldown:** Automatically flags donors in cooldown rest periods to prevent unsafe donations.
- **Direct Call & WhatsApp Action:** Instant contact initiation without friction.

B. Emergency Blood Request Workflow:

- **Urgency Priority Levels:** Routine (48-72 hrs), Urgent (12-24 hrs), and Critical (<2 hrs).
- **Auto-Expiration Daemon:** Background checks auto-expire stale requests to keep the feed fresh.
- **Progressive Units Tracking:** Visual bar showing units fulfilled vs units needed.
- **Pledge & Verification Lifecycle:** Donor RSVP -> Live GPS Share -> Hospital/Patient mark as Received.

C. Hospital Command Center & Blood Bank:

- **Stock Reserves Module:** Real-time inventory tracking for all blood groups with critical threshold badges.
- **Predictive Dengue Season Analytics:** AI/ML model alerting hospitals of upcoming regional platelet shortages.
- **Hospital Requests Dashboard:** View and verify blood bags received for admitted patients.

D. User Experience, PWA & Security:

- **QR Code Digital Health Card:** Base64 QR code with donor history, badge, and blood group for instant hospital verification.
- **Progressive Web App (PWA):** Installable on Android & iOS mobile devices with offline service worker caching.
- **Role-Based Authentication:** Dedicated roles for Donors, Patients, and Hospitals secured via JWT tokens.

4. How The Project Works (Step-by-Step Workflow)

Step 1: User & Donor Onboarding

A donor registers on /auth with their email, city, blood group, and phone number. Their profile is created with eligibility checks, cooldown dates, and an initial donor badge.

Step 2: Emergency Request Creation

A patient or hospital creates an emergency request on /request specifying patient name, hospital, city, blood group, units, and urgency. Django signals auto-trigger email alerts and mock social media broadcasts to matching local donors.

Step 3: Discovery & Donor Matching

Users locate donors on /search by searching cities or blood groups. The priority algorithm sorts donors by physical proximity, donation recency, and response rating.

Step 4: Real-time Coordination & Live Tracking

The requester contacts the donor directly via Phone, 1-click WhatsApp, or Daphne WebSocket chat. When the donor clicks 'I will donate', they can activate 'Share GPS Location', sending live coordinates to the hospital/patient.

Step 5: Donation Verification & Gamification

Once the donation is completed at the hospital, the patient or hospital marks it as 'Received'. The donor's total donation count increases, badges upgrade (Bronze -> Silver -> Gold), and an official Digital Certificate of Appreciation is generated.

Step 6: Hospital Blood Bank Management

Hospital administrators manage their facility's blood inventory via the Hospital Command Center, adjusting available units and receiving automated low-stock warnings.

5. Local Execution & Credentials

Frontend: Run npm run dev inside LifeDrop/ (Runs at http://localhost:5173).

Backend: Run python manage.py runserver inside LifeDrop/backend/ (Runs at http://127.0.0.1:8000/api/).

Demo Accounts:

- Hospital Admin: admin@lifestream.com / Password: Admin@1234
- Standard Donors: salman@gmail.com, ahmad@gmail.com, amna@gmail.com / Password: password123